

- As water is lost by the leaves through transpiration, the air bubble moves along the graduated tube.

**(a) (i)** State the independent variable **and** the dependent variable in this investigation.

*dependent* .....

(ii) List **three** variables that should be controlled in this investigation **and** describe how the student could standardise **two** of these variables.

[3]

[3]



- (c) The student carried out further experiments, using the same apparatus, to investigate the effect of two different environmental carbon dioxide concentrations on the rate of transpiration. These experiments were carried out at a high light intensity and at a low light intensity.

The leafy shoots used in the experiments were taken from the same plant and each shoot had five leaves.

The student calculated the percentage reduction in the transpiration rate from 50 ppm to 730 ppm of carbon dioxide at low light intensities.

The results are shown in Table 1.1.

**Table 1.1**

| concentration of carbon dioxide / ppm | light intensity | transpiration rate / $\text{g dm}^{-3} \text{ hr}^{-1}$ | percentage reduction in transpiration rate from 50 ppm to 730 ppm of carbon dioxide |
|---------------------------------------|-----------------|---------------------------------------------------------|-------------------------------------------------------------------------------------|
| 50                                    | low             | 1.28                                                    | 36.7                                                                                |
| 730                                   | low             | 0.81                                                    |                                                                                     |
| 50                                    | high            | 3.03                                                    |                                                                                     |
| 730                                   | high            | 2.12                                                    |                                                                                     |

ppm = parts per million

- (i) Complete Table 1.1 by calculating the percentage reduction in transpiration rate from 50 ppm to 730 ppm of carbon dioxide at the **high light intensity**.

Show your working.

- (ii) The student concluded that, as carbon dioxide concentration increased, the transpiration rate in plants decreased at all light intensities.

Explain why this conclusion may **not** be valid.

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.....[2]

[Total: 15]